



Lab 9 (Mini-Project 1)
Two Stage Miller OTA Design

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Part 1: g_m/I_D Design Charts

Using ADT Device Xplore, plot the following design charts vs g_m/I_D for both PMOS and NMOS. Set $V_{DS} = V_{DD}/3$ and $L = 0.28\mu, 0.5\mu, 1\mu, 1.5\mu, 2\mu$.

- g_m/g_{ds}
- I_D/W
- g_m/C_{gg}
- V_{GS}

For NMOS:

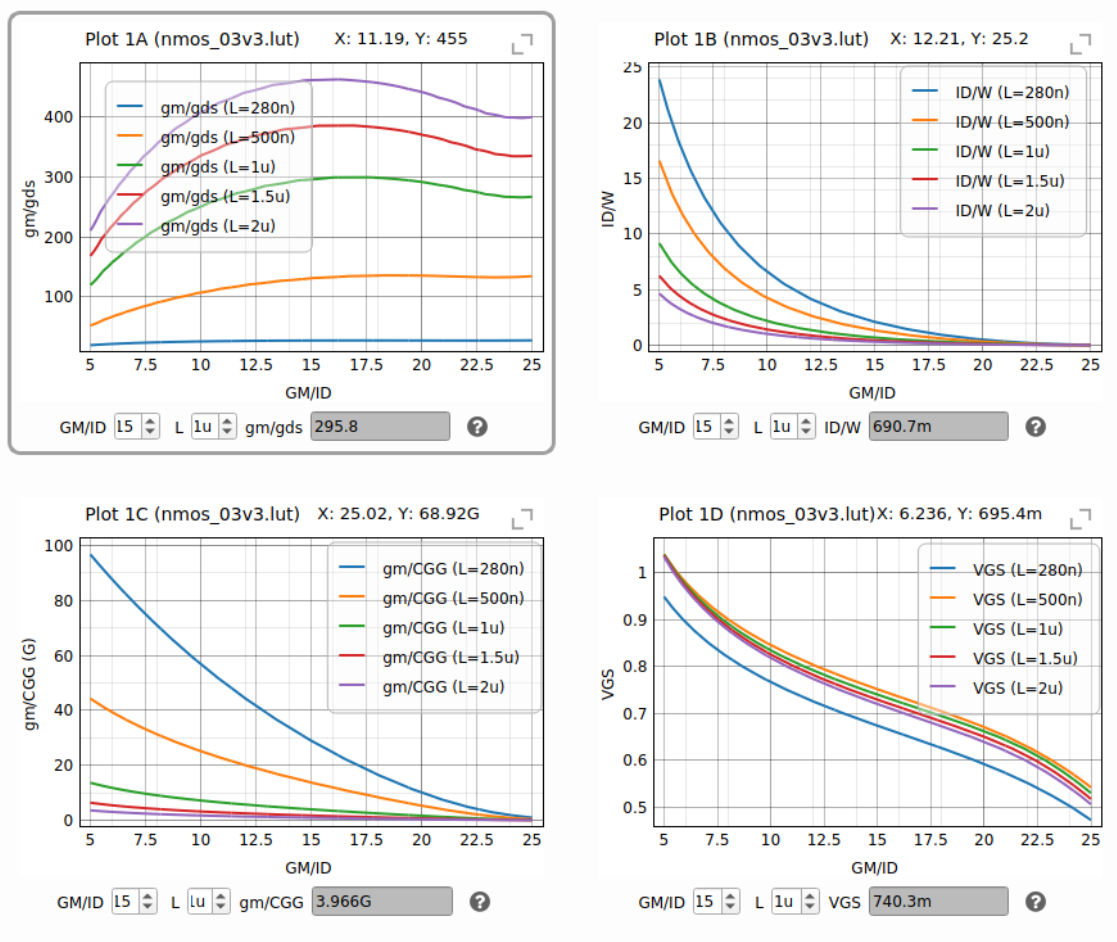


Fig.1: NMOS Design Charts

Part 1: gm/ID Design Charts

- g_m/g_{ds}
- I_D/W
- g_m/C_{gg}
- V_{GS}

For PMOS:

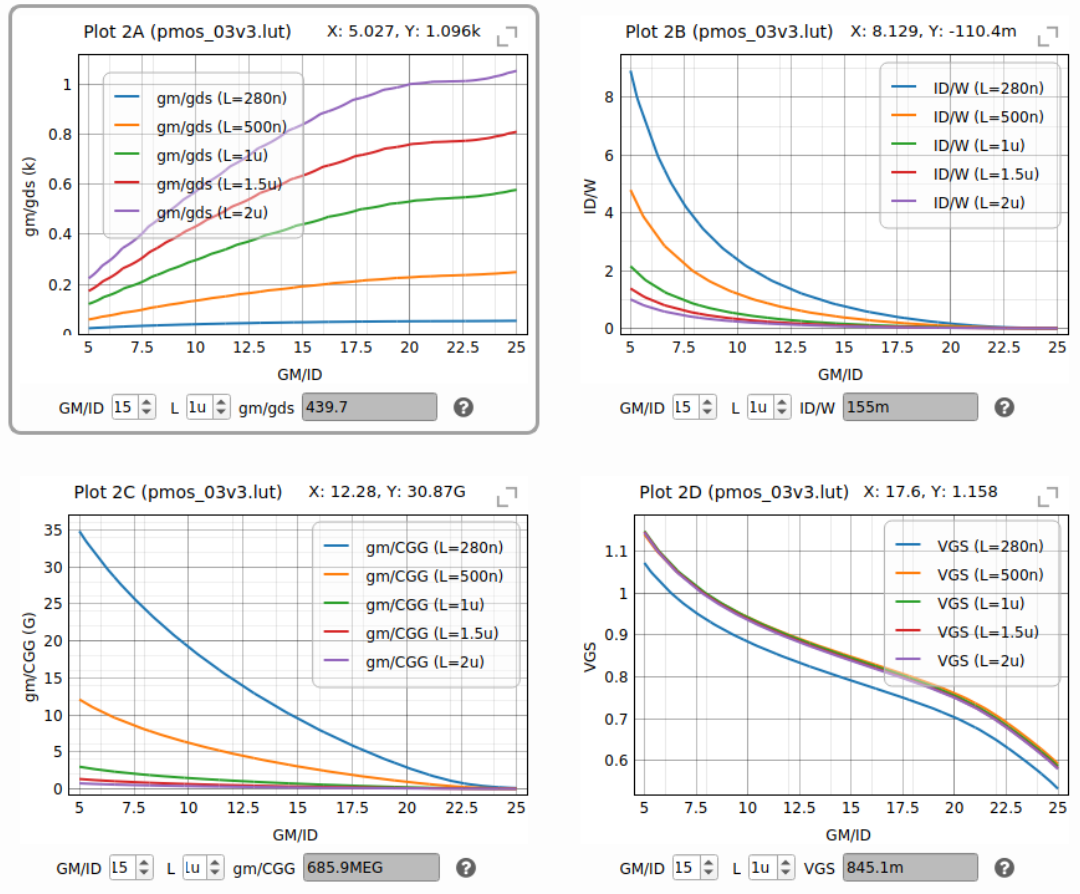


Fig.2: PMOS Design Charts

Part 2: OTA Design

Required Specs.

Use g_m/I_D methodology to design a differential input, single-ended output **two-stage Miller-compensated OTA**. The OTA is to be used as a buffer (unity gain feedback configuration) to probe sensitive internal signals in a complex mixed-signal design. The OTA should achieve the specs below:

Parameter	Values
Technology	0.18 μ
Supply Voltage V_{DD}	1.8V
Static Gain Error ϵ_r	$\leq 0.05\%$
CMRR @ DC	≥ 74 dB
Phase Margin (avoid pole-zero doublets)	$\geq 70^\circ$
OTA Current Consumption	≤ 60 μ A
CMIR – high	≥ 0.8 V
CMIR – low	≤ 0.2
Output Swing	0.2 – 1.6V
Load C_L	5pF
Buffer Closed Loop rise time t_{rise}	≤ 70 ns
Slew Rate SR	5 V/ μ s

Table 1: Desired Specs

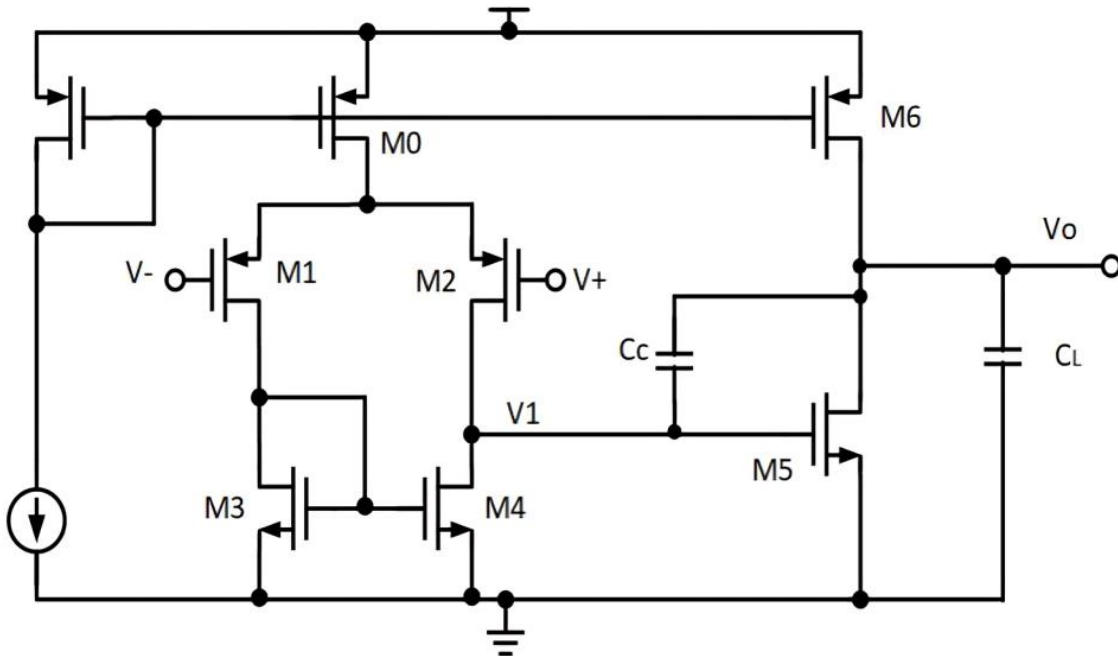


Fig.3: Two Stage Miller OTA Schematic

Topology Selection

- From required specs we conclude the following:

- We need high Open-Loop gain A_{vOL}

$$\text{Static Gain error } \epsilon_r = \frac{1}{1+LG} = \frac{1}{1+\beta A_{OL}} = 0.05\% \quad (\text{note that } \beta = 1)$$

$$A_{OL} \geq 2000$$

$$A_{OL} \geq 66 \text{ dB}$$

- We need Wide Output Swing $0.2 - 1.6V$

Based on these explanations we can use Two-Stage Miller OTA to achieve the required specs.

- CMIR is close to ground $0.2 - 0.8V$
 - We Can achieve that using PMOS input pair in the first Stage.

Pre-Sizing

- We assume a reasonable starting point for C_C

$$C_C = 0.5 \times C_L = 2.5pF$$

- Using t_{rise} requirements, we can calculate UGF and $g_{m2,3}$

$$\tau = \frac{t_{rise}}{2.2} = 31.81ns$$

$$UGF = \frac{1}{\tau} = \frac{g_{m2,3}}{C_C}$$

$$g_{m2,3} = 78.57\mu S$$

- Using Slew Rate SR requirements, we can calculate the required current in the first stage

$$SR = \frac{I_{B1}}{C_C}$$

Taking Stability into account we can say

$$I_{B1} = 12.5\mu A$$

$$I_{B2} = 60 - 12.5 = 47.5\mu A$$

$$I_{B2} \leq 47.5\mu A$$

- We have to make the output pole of the first stage dominant and far away from second stage output pole for stability considerations

So, we can assume first stage gain is twice of second stage (6 dB difference)

$$A_{OL1} = 36dB$$

$$A_{OL2} = 30dB$$

Part 2: OTA Design

- Calculate the g_m/I_D of the first stage

$$\frac{g_m}{I_D} = \frac{78.57\mu}{6.25\mu} = 12.57 \text{ S/A}$$

- Show that Closed-Loop gain is a buffer

$$A_{vCL} \approx 1 - \frac{1}{A_{vOL}} = 1 - \frac{1}{2000} = 0.9995$$

- Calculate the required DC gain

$$A_{vd} = 66 \text{ dB}$$

- Verify that your g_m/I_D choices do not violate the CMIR and the peak-to-peak output swing

$$\frac{g_m}{I_D} \approx \frac{2}{V_{Dsat}}$$

$$V_{inCM_{max}} \leq V_{DD} - |V_{DS_{sat1}}| - |V_{GS3}|$$

$$V_{inCM_{max}} \approx 0.78$$

$$V_{inCM_{min}} \geq V_{GS6} - |V_{GS3}| + |V_{DS_{sat3}}|$$

$$V_{inCM_{min}} \approx 0.022$$

$$V_{out_{max}} = V_{DD} - V_{Dsat4}$$

$$V_{out_{max}} = 1.68V$$

$$V_{out_{min}} = V_{Dsat7}$$

$$V_{out_{min}} = 122.8mV$$

- Implementation of R_z , we placed this resistance for biasing and we match its value to M_4 resistance $\left(R_z \approx \frac{1}{g_{m4}}\right)$, $g_{m4} = 628.75 \mu S$

$$R_z \approx 1.6k\Omega$$

OTA Devices Sizing

Parameters	$M_{2,3}$ Input Pair	$M_{5,6}$ CM load	M_1 tail CS	M_0 tail CM	M_4 2 nd Load	M_7 2 nd Input Stage
W	$9\mu\text{m}$	$5.28\mu\text{m}$	$25.76\mu\text{m}$	$19.2\mu\text{m}$	$92\mu\text{m}$	$17.44\mu\text{m}$
L	440 nm	560 nm	380 nm	380 nm	380 nm	363 nm
g_m	$78.57\mu\text{S}$	$93.72\mu\text{S}$	$187.5\mu\text{S}$	$147.4\mu\text{S}$	$712\mu\text{S}$	$615.5\mu\text{S}$
g_m/I_D	12.57	15	15.2	15	15.2	13
$V_{D_{sat}}$	137.5mV	108.9mV	109.1mV	111.5mV	109.2mV	122.8mV
V_{ov}	105mV	43.9mV	56.33mV	62.22mV	58.5mV	66.22mV
V^*	161.5mV	133.4mV	132mV	135.6mV	133.4mV	154.3mV

Table 2: OTA Devices Sizing

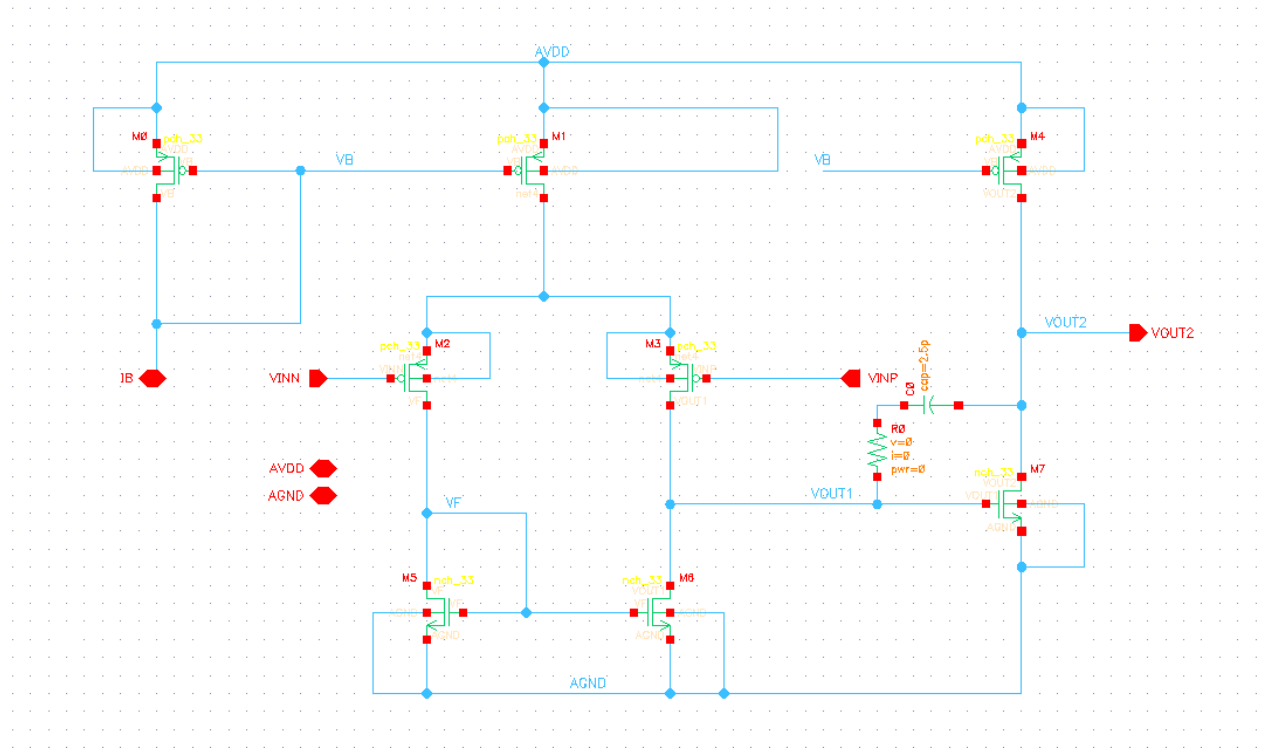


Fig.4: Two Stage OTA Schematic

Part 3: Open-Loop OTA Simulation

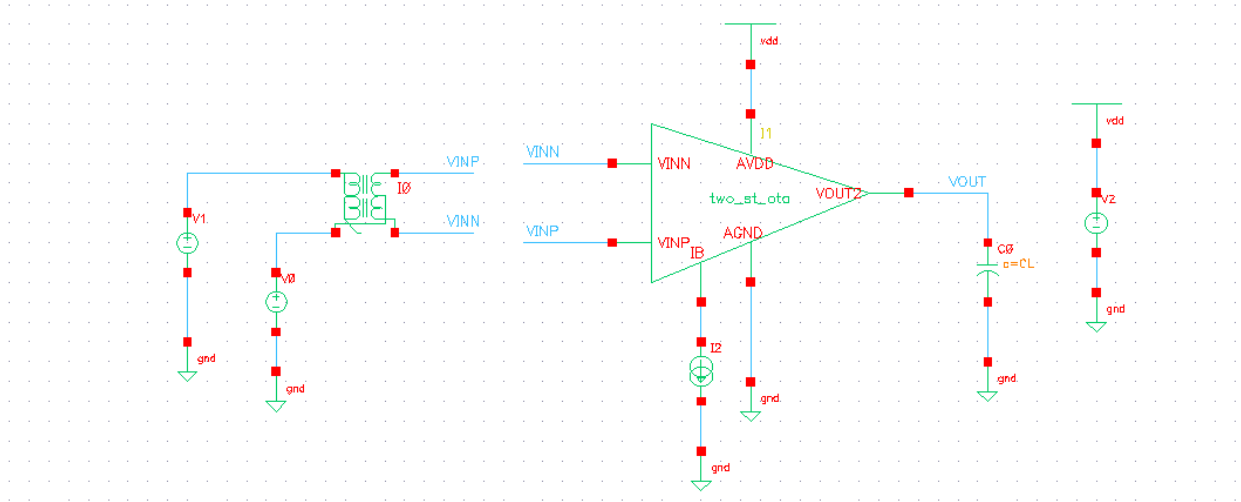


Fig.5: Open-Loop OTA testbench

OP and Biasing Voltages

Schematic of the OTA and bias circuit with DC node voltages clearly annotated.



Fig.6: Open-Loop OTA DC OP

Part 3: Open-Loop OTA Simulation

- Is the Current and g_m in the input pair exactly equal?

Yes, Current and transconductance are exactly equal, because there is no mismatch between the PMOS input pair.

- What is the DC Voltage at the output of first Stage? Why?

$$V_{out_1} = 748.2mV$$

Because the output node of the first stage follow the mirror node of the NMOS CM Load.

- What is the DC Voltage at the output of Second Stage? Why?

$$V_{out_2} = 953.5 mV$$

This dc level of V_{out_2} as it's high impedance node so it's ill-defined, but we recovered that problem by taking V_{DS} into account by putting a spec on V_{DS} by $0.9V$.

Diff. Small Signal Ccs

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade) $V_{in.diff_{AC}} = 1V, V_{in_{CM}} = 0.6V$

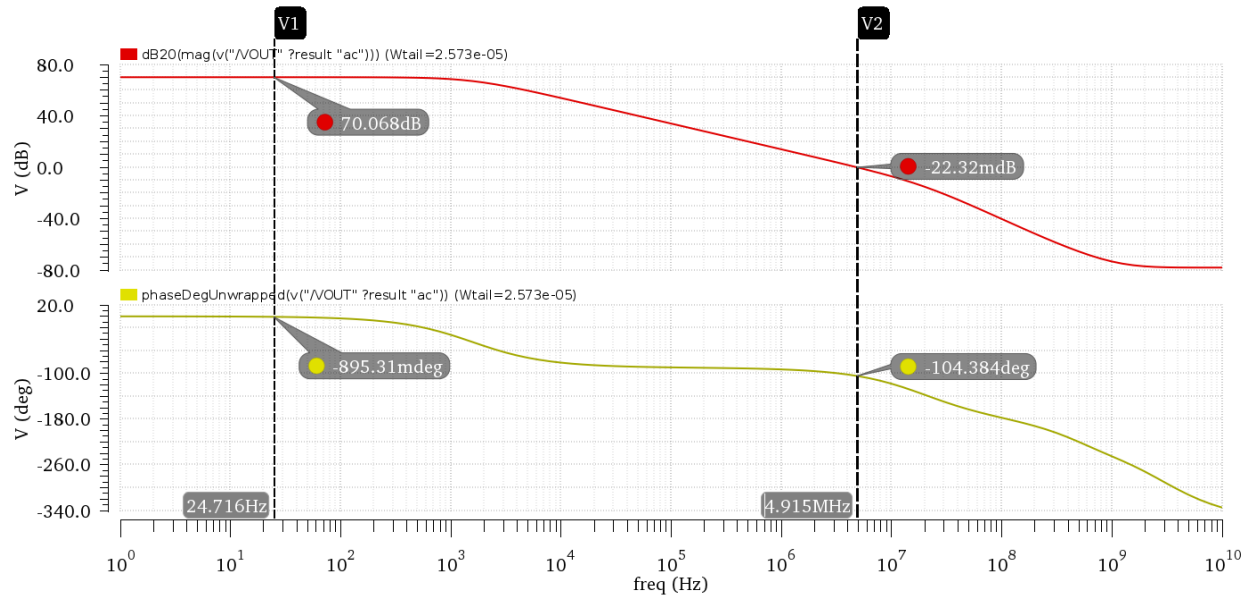


Fig.7: Open-Loop OTA Bode Plot of diff. gain mag. and phase

- Hand Analysis

$$A_{vd_1} = g_{m2,3} R_{out_1} = g_{m2,3}(r_{o3}/r_{o6})$$

$$A_{vd_2} = g_{m7} R_{out_2} = g_{m7}(r_{o4}/r_{o7})$$

$$A_{vd} = A_{vd_1} \times A_{vd_2}$$

$$\omega_{p1} \approx \frac{1}{R_{out_1}(G_{m_2}R_{out_2})C_C}, BW = \frac{\omega_{p1}}{2\pi}$$

Part 3: Open-Loop OTA Simulation

$$GBW = \frac{g_{m2,3}}{C_c}$$

$$UGF = \omega_u \approx GBW = A_{vd} \times BW$$

Parameter	Analytical	Simulation
Diff. gain A_{vd}	3.196k	3.188k
Diff. gain A_{vd} dB	70.1 dB	70.07 dB
BW	1.637 kHz	1.578 kHz
GBW	5.23 MHz	5.042 MHz
UGF	5.23 MHz	5.031 MHz

Output	Nominal	Spec	Weight	Pass/Fail
dB20(mag(v("/VOOUT" ?result ...				
phaseDegUnwrapped(v("/VO...				
Ao	3.188k	> 2k		pass
Ao_dB	70.07 dB	> 66		pass
BW	1.578k			
GBW	5.042 MHz	> 5M		pass
Fu	5.031 MHz	> 5M		pass
CMRR				
CMRR dB	75.74 dB	> 74		pass
Vout1 dc	748.2 mV			
Vout2 dc	953.5 mV			

Fig.8: Diff. Simulation Results

CM Small Signal Ccs

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade) $V_{inCMAC} = 1V$, $V_{inCM} = 0.6V$

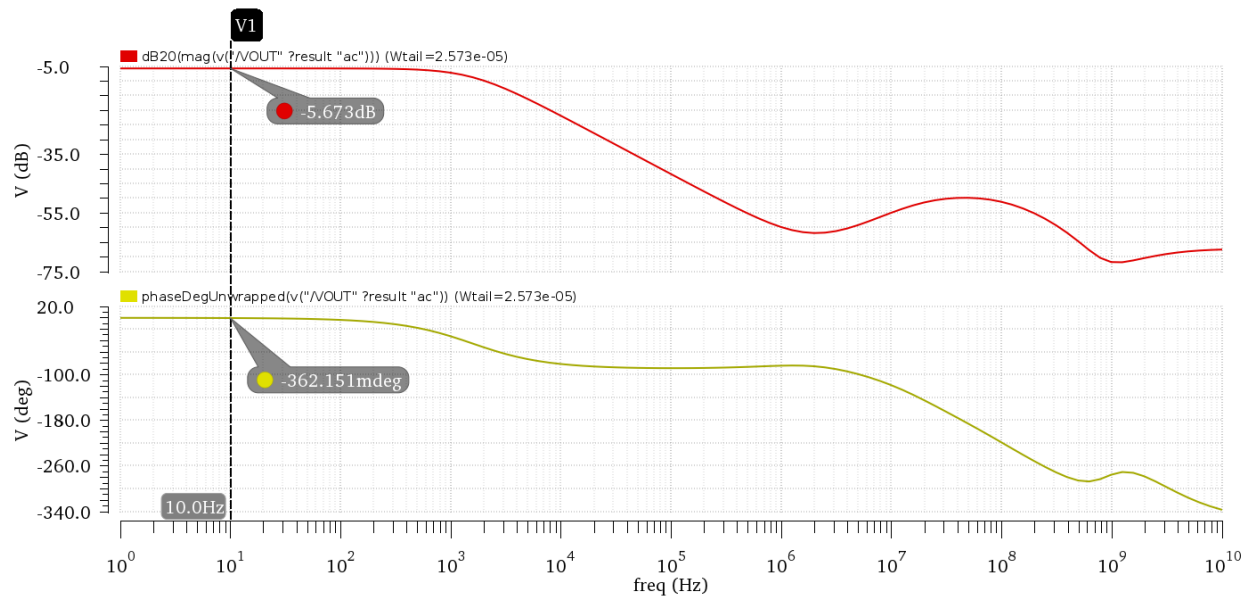


Fig.9: Open-Loop OTA Bode Plot of CM gain mag. and phase

Part 3: Open-Loop OTA Simulation

- Hand Analysis

$$A_{vCM_1} = \frac{1}{2g_{m5,6}R_{ss}} = \frac{g_{ds1}}{2g_{m5,6}}$$

$$A_{vCM} = A_{vCM_1} \times A_{vd_2}$$

Parameter	Analytical	Simulation
CM gain A_{vCM}	534 m	520.4 m
CM gain A_{vCM} dB	-5.44 dB	-5.673 dB
CMRR	75.54 dB	75.74 dB

Ao	520.4m	< 600		pass
Ao_dB	-5.673 dB	> -7.97		pass
CMRR				
CMRR dB	75.74 dB	> 74		pass

Fig.10: CM Simulation Results

CMRR

$$CMRR = \frac{A_{vd}}{A_{vCM}}$$

$$CMRR_{dB} = A_{vd_{dB}} - A_{vCM_{dB}}$$

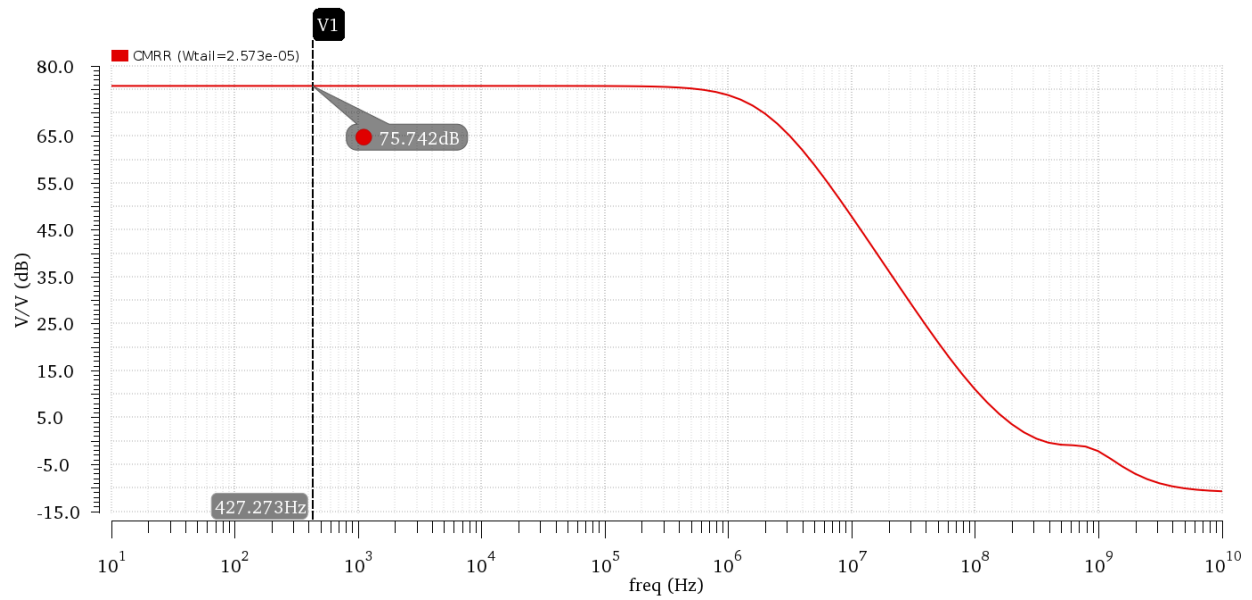
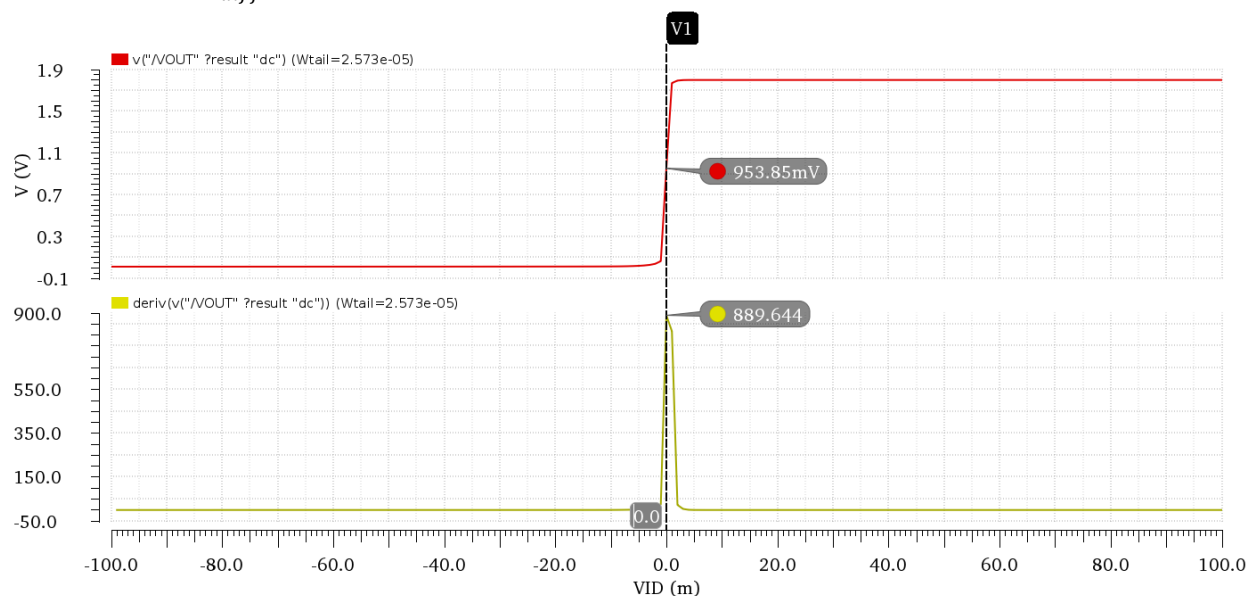


Fig.11: Bode Plot of CMRR

Diff. Large Signal Ccs

Use DC Sweep $V_{in_diff} = -0.1: 1m: 0.1$, we used very small step because the gain region is very small

Fig.12: Output Voltage & its derivative vs V_{id}

- What is the Value of $V_{out}@V_{in_diff} = 0$?

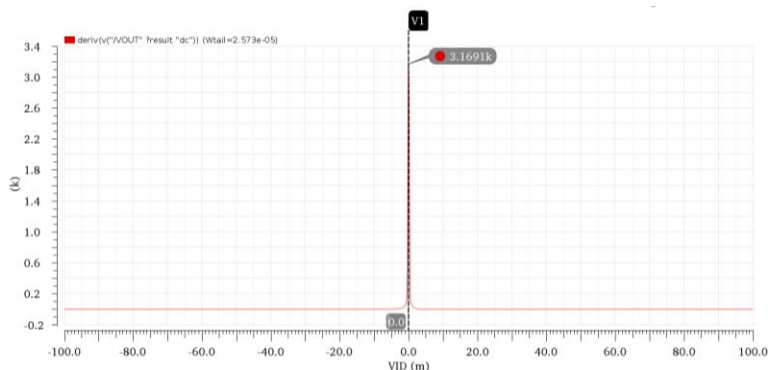
Parameter	Diff. LS Ccs	OP
DC Output Voltage V_{out}	953.85mV	953.5mV

Note that $V_{out}@ (V_{in_diff} = 0)$ is equal to V_{out} (obtained from dcOP)

- From the derivative of V_{out} vs V_{in_diff} . Is the peak less than the value of A_{vd} obtained from ac analysis? Why?

Yes, the peak is less than actual gain, but if we take a small step in dc analysis the results will be a bit accurate.

like here I take a step of (0.1mV) instead of (1mV) in the previous simulation.

Fig.13: Output Voltage derivative vs V_{id}

CM Large Signal Ccs (region vs V_{inCM})

Use DC Sweep $V_{inCM} = 0: 1m: V_{DD}$

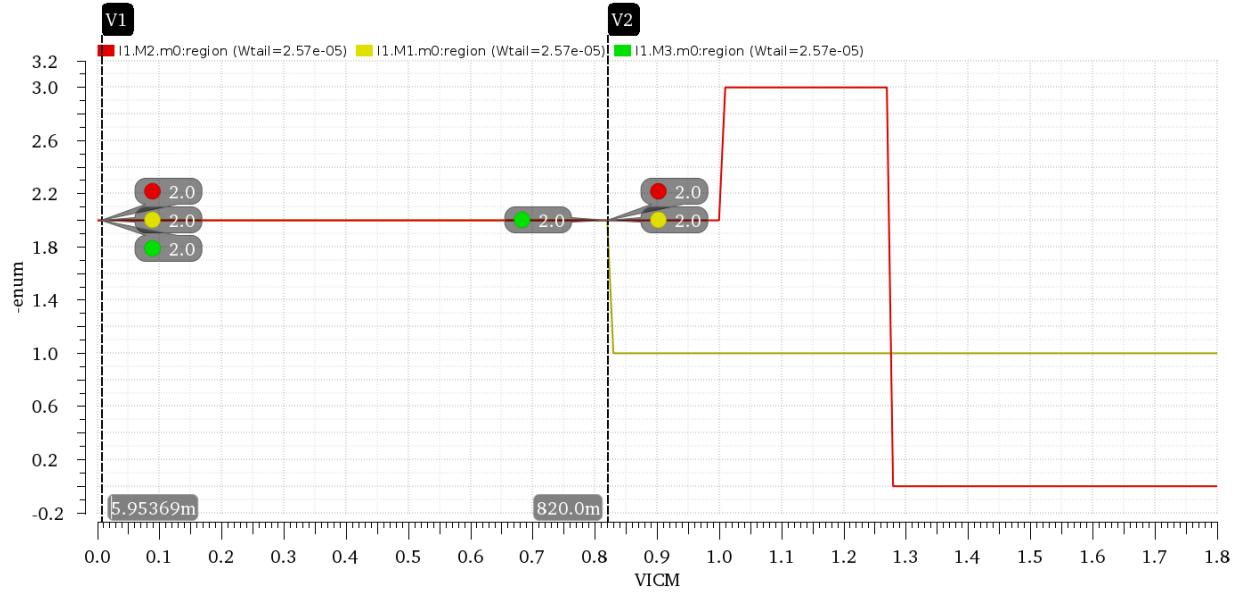


Fig.14: region vs V_{inCM}

$$0 \leq CMIR \leq 820mV$$

- Hand Analysis

$$V_{inCM_{max}} \leq V_{DD} - |V_{DS_{sat1}}| - |V_{GS3}|$$

$$V_{inCM_{max}} \approx 863.5mV$$

$$V_{inCM_{min}} \geq V_{GS6} - |V_{GS3}| + |V_{DS_{sat3}}|$$

$$V_{inCM_{min}} \approx -36.4mV$$

Parameter	Diff. LS Ccs	OP
$CMIR_{Low}$	0	-36.4 mV
$CMIR_{High}$	820mV	863.5 mV

CM Large Signal Ccs (GBW vs V_{inCM})

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade). $V_{in_{diff.AC}} = 1$ and $V_{in_{CM.AC}} = 0$

Use Parametric Sweep $V_{in_{CM}} = 0:25m:V_{DD}$

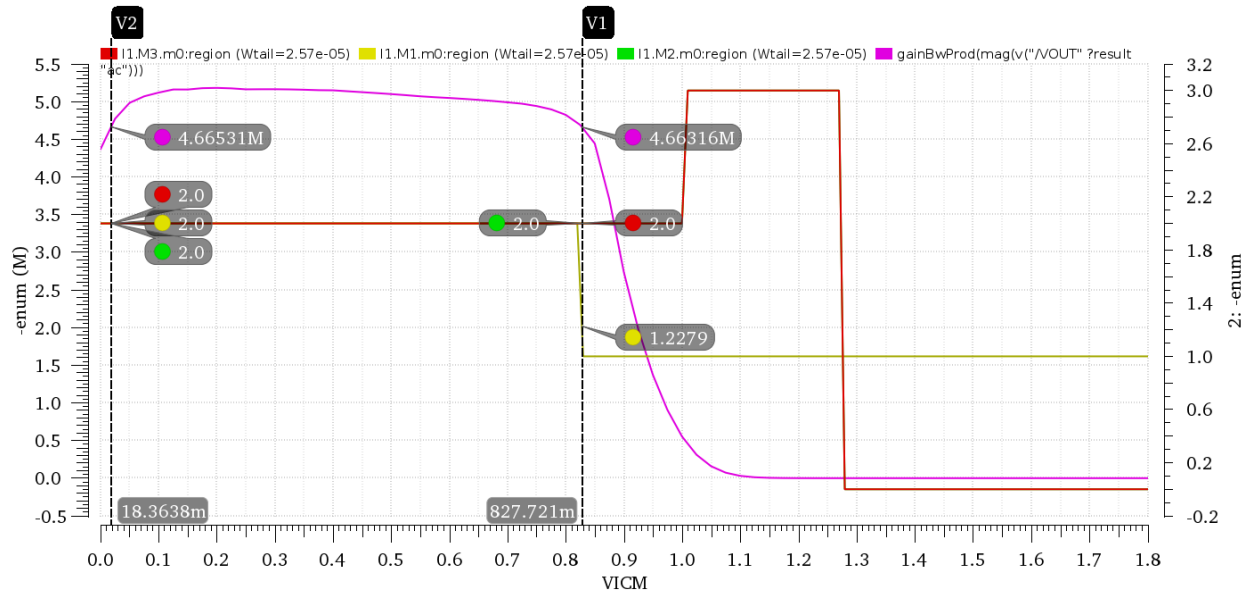


Fig.15: GBW vs V_{inCM}

$$GBW_{max} = 5.18 \text{ MHz}$$

$$90\% \text{ of } GBW = 4.66 \text{ MHz}$$

Parameter	region	OP	GBW
$CMIR_{Low}$	0	-36.4 mV	18.36 mV
$CMIR_{High}$	820 mV	863.5 mV	827.72 mV

We Noticed that the CMIR range extends to ground, due to body effect. In CMOS fabrication process PMOS is placed in n-well, so we can connect the source with body without any problem, source voltage drops within some range this affects the V_{SB} and make it decrease then the $|V_{THp}|$ magnitude increase so it pushes the minimum required gate voltage boundary lower and lower, effectively allowing CMIR to stretch all the way down to ground.

Part 4: Closed-Loop OTA Simulation

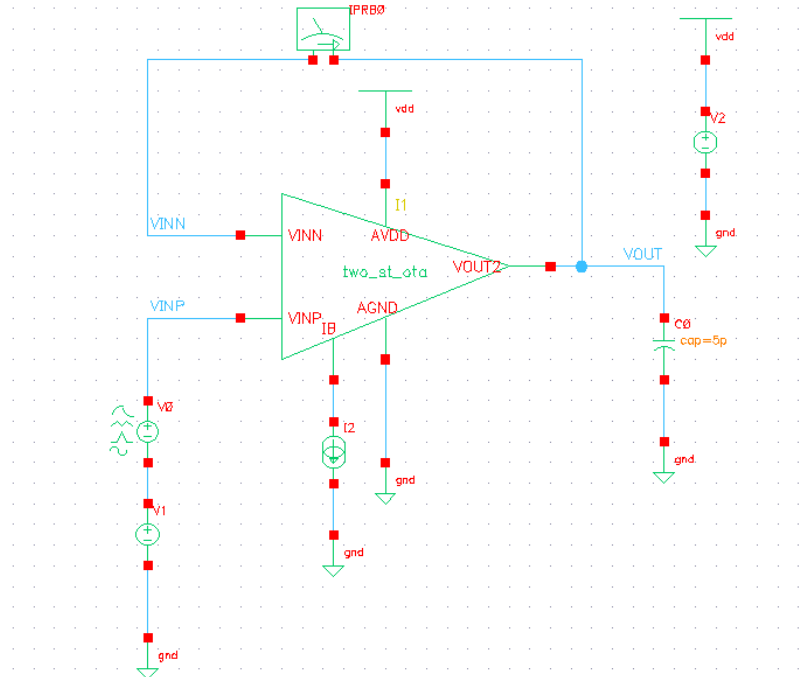


Fig.16: Closed-Loop OTA testbench

OP and Bias Voltages

Schematic of the OTA and bias circuit with DC node voltages clearly annotated.

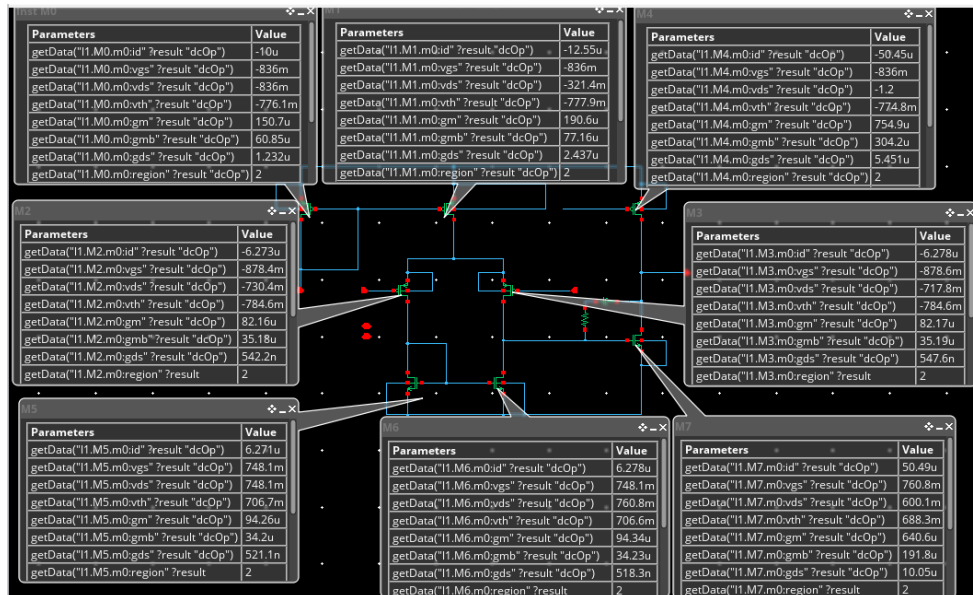


Fig.17: Closed-Loop OTA DC OP

Part 4: Closed-Loop OTA Simulation

- Are the DC voltages of the op-amp terminals exactly equal? Why?

Yes, V_{in+} and V_{in-} are exactly equal, due to the massive A_{vOL} and negative feedback minimizes the error signal.

$$V_{err} = \frac{V_{in+}}{1 + \beta A_{vOL}} = \frac{V_{in+}}{1 + LG}$$

$$V_{err} = 0.188mV$$

$$V_{out} = A_{vOL}(V_{in+} - V_{in-})$$

$$V_{out} = V_{in-}$$

$$V_{in-} = V_{in+} \left(\frac{A_{vOL}}{1 + A_{vOL}} \right)$$

$$V_{in-} \approx V_{in+}$$

$$\epsilon_r = \frac{1}{1 + LG} = 0.03\%$$

- Is the DC Voltage at the output of first Stage equal to the value in the open-loop simulation? Why?

$$V_{out_1} = 756.9mV$$

Because the V_{out_2} decreased. So, V_{out_1} must increase to keep the same current in second stage

- Is the Current and g_m in the input pair exactly equal?

As V_{out_1} increased M_2 get a higher V_{DS} which made a slight difference between M_2 and M_3 .

Vout1 dc	756.9m
Vout2 dc	600.1m
Vin -	600.1m
Vin+	600m
id2	-6.272u
id3	-6.277u
gm2	82.14u
gm3	82.17u

Fig.18: Closed-Loop OTA DC OP results

Loop Gain (STB Analysis)

Use STB analysis (1Hz:10Gz, logarithmic, 10 points/decade) $V_{in_{CM}} = 0.6V$

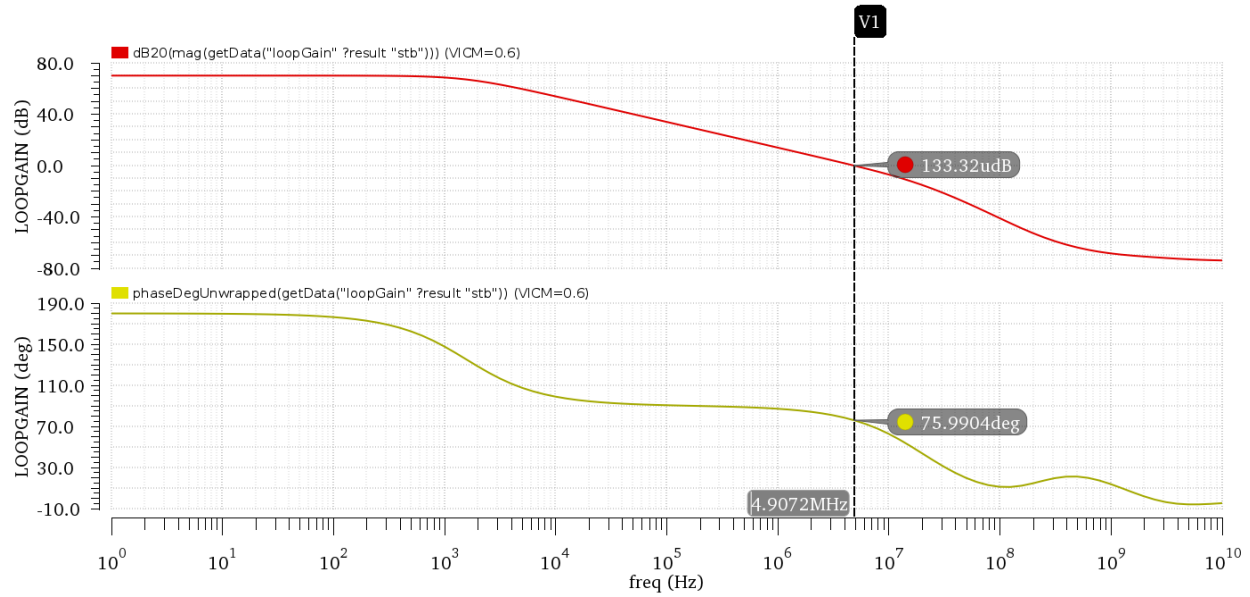


Fig.19: Bode Plot of LG mag. and phase

Parameter	CL Simulation	OL Simulation
DC gain A_{vd}	3.074k	3.188k
DC gain A_{vd} dB	69.75 dB	70.07 dB
GBW	5.049 MHz	5.042 MHz
UGF	5.036 MHz	5.031 MHz

The specs vary between open and closed loop simulations due to $V_{out_{dc}}$ is changed so V_F too and there is error signal at the input therefore OP parameters are changed.

- Hand Analysis

$$A_{vd} = g_{m2,3}(r_{o3}/r_{o6}) \times g_{m7}(r_{o4}/r_{o7})$$

$$\omega_{p1} \approx \frac{1}{R_{out1}(G_{m2}R_{out2})C_c}, \quad BW = \frac{\omega_{p1}}{2\pi}$$

$$GBW = \frac{g_{m2,3}}{C_c}$$

$$UGF = \omega_u \approx GBW = A_{vd} \times BW$$

$$PM = 90^\circ - \tan^{-1}\left(\frac{\omega_u}{\omega_{p2}}\right), \quad \frac{\omega_u}{\omega_{p2}} = \frac{g_{m2,3}}{C_c} \times \frac{C_L}{g_{m7}}$$

Part 4: Closed-Loop OTA Simulation

<i>Parameter</i>	<i>Analytical</i>	<i>CL Simulation</i>
<i>DC gain A_{vd}</i>	<i>3.17k</i>	<i>3.074k</i>
<i>DC gain A_{vd} dB</i>	<i>70.02 dB</i>	<i>69.75 dB</i>
<i>BW</i>	<i>1.646 kHz</i>	<i>1.638 kHz</i>
<i>GBW</i>	<i>5.22 MHz</i>	<i>5.049 MHz</i>
<i>PM</i>	<i>75.61°</i>	<i>76.14°</i>

BW, GBW, & PM hand analysis are different from the simulation as the parasitic caps are large due to large sizes and they are not taken into consideration in hand analysis.

LG	3.074k			
LG dB	69.75			
BW	1.638k			
GBW	5.049M			
Fu	5.036M			
PM	76.14	> 70		pass
GM	71.1			
Vout1 dc	756.9m			
Vout2 dc	600.1m			
Vin -	600.1m			
Vin+	600m			
id2	-6.272u			
id3	-6.277u			
gm2	82.14u			
gm3	82.17u			

Fig.20: STB Analysis Results

Part 4: Closed-Loop OTA Simulation

Slew Rate

Apply a step input with the following parameters (delay = 1 μ s, initial value = CMIR-low + 50mV, final value = CMIR-high - 50mV, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step input, which is why we selected very large period and width for the pulse.

Run transient analysis (stop = 5 μ s and step = 0.1ns).

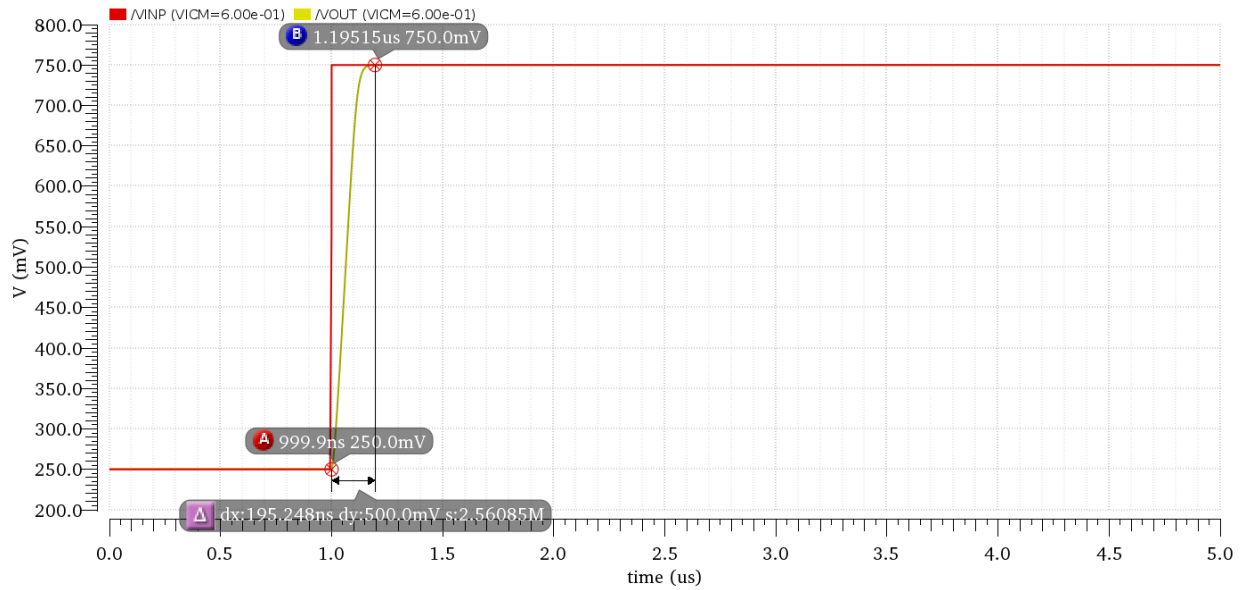


Fig.21: Input and Output Step Response

Slew Rate	4.332	> 5	fail
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Fig.22: Slew Rate Result

We Notice that slew rate didn't meet the specs so we make random sweeps on C_C to determine a reasonable value of it.

($C_C = 0.5 \times C_L = 2.5pF$ was a starting point and we may refine this value with some sweeps)

I made a Parametric sweep on C_C from 2p to 2.5p with step 0.1p, and $C_C = 2.1pF$ satisfies the spec on slew rate, then I recalculated the phase margin to ensure it satisfies the spec of ($\geq 70^\circ$) with the new value of C_C .

LG	1.44k			
LG dB	63.17			
BW	4.191k			
GBW	6.049M			
Fu	6.034M			
PM	73.84	> 70		pass
GM	70.54			
Slew Rate	5.124	> 5		pass

Fig.23: Slew Rate Result after Modification

Part 4: Closed-Loop OTA Simulation

$$SR \approx \frac{I_{B1}}{C_c}$$

Parameter	Analytical	Simulation
Slew Rate SR	5.05 V/ μ s	5.124 V/ μ s

The Hand Analysis is smaller due to parasitic capacitance neglection in calculation.

Settling Time

Apply a step input with the following parameters (delay = 1 μ s, initial value = $V_{DD}/3$, final value = $V_{DD}/3$, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step input, which is why we selected very large period and width for the pulse. Note that we apply a small signal pulse (5mV step) to measure the small signal settling time.

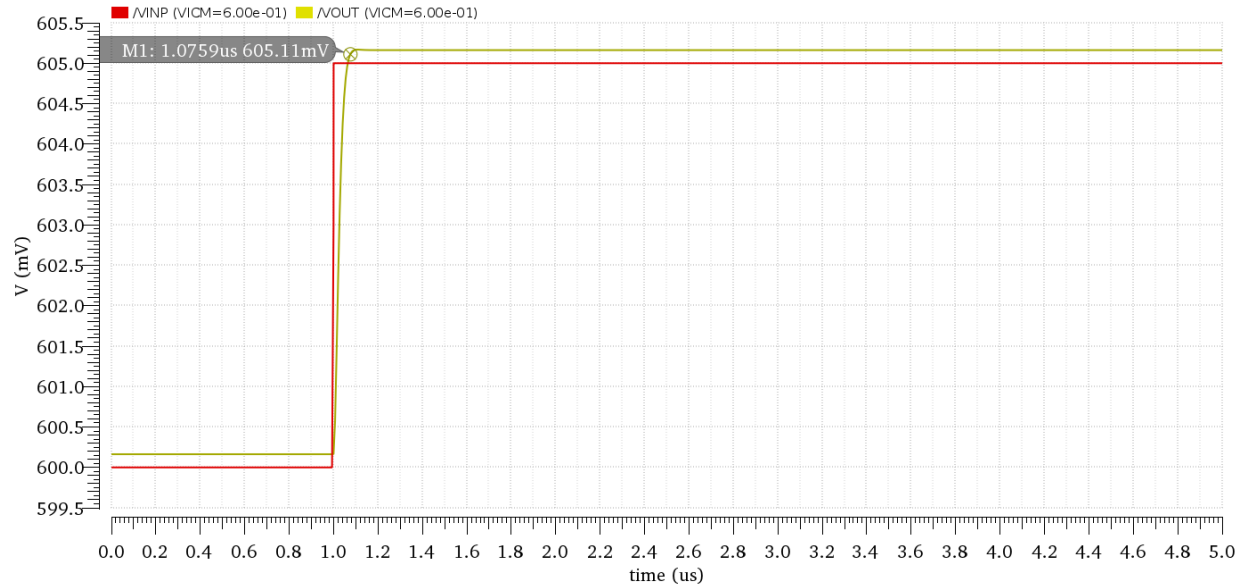


Fig.24: Input and Output Step Response

LG	3.178k			
LG dB	70.04			
BW	1.886k			
GBW	6.01M			
Fu	5.996M			
PM	73.6	> 70		pass
GM	71.24			
trise	39.8n	< 70		pass
tail CS	-12.55u			

Fig.29: Settling time results

Part 4: Closed-Loop OTA Simulation

$$t_{rise} = 2.2 \times \tau$$

$$UGF = \frac{1}{\tau} = \frac{g_{m2,3}}{C_C}$$

$$t_{rise} = 54.9 ns$$

Parameter	Analytical	Simulation
Rise time t_{rise}	54.9 ns	39.8 ns

The simulation result is better as hand analysis approximates the system to a 1st order system but in reality, it's a 2nd order system which has faster response.

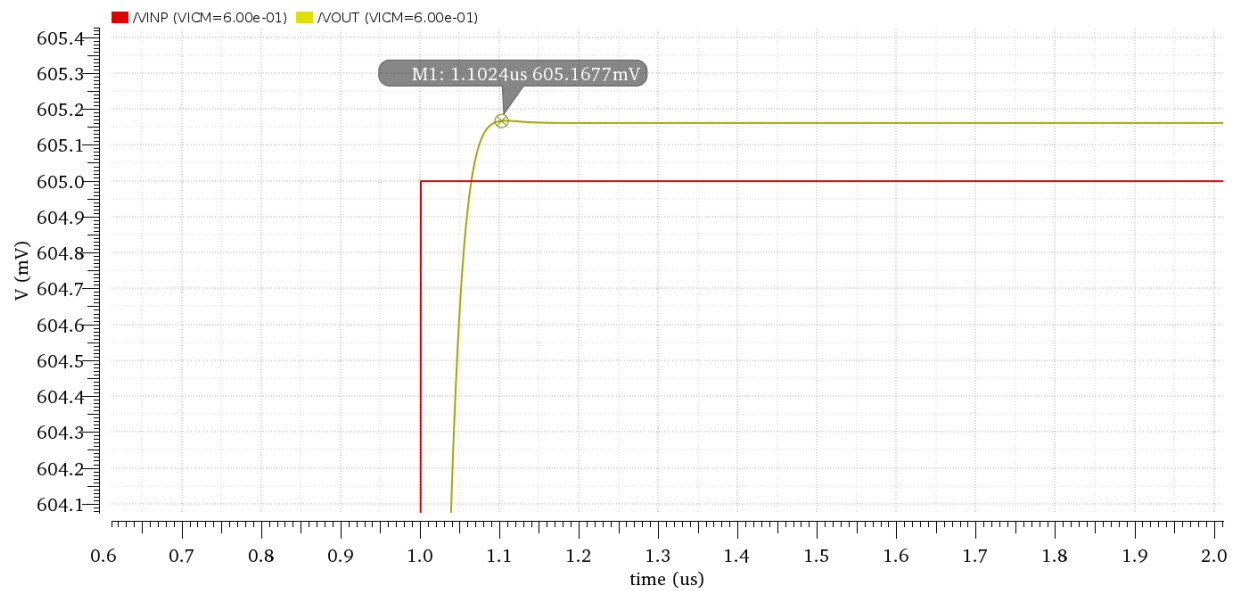


Fig.25: Close look of Input and Output Step Response

- Do you see any ringing? Why?

We notice a small ringing behaviour in the output signal, because $PM < 76^\circ$ so it's underdamped.

Part 5: DC Closed-Loop AC Open-Loop OTA Simulation

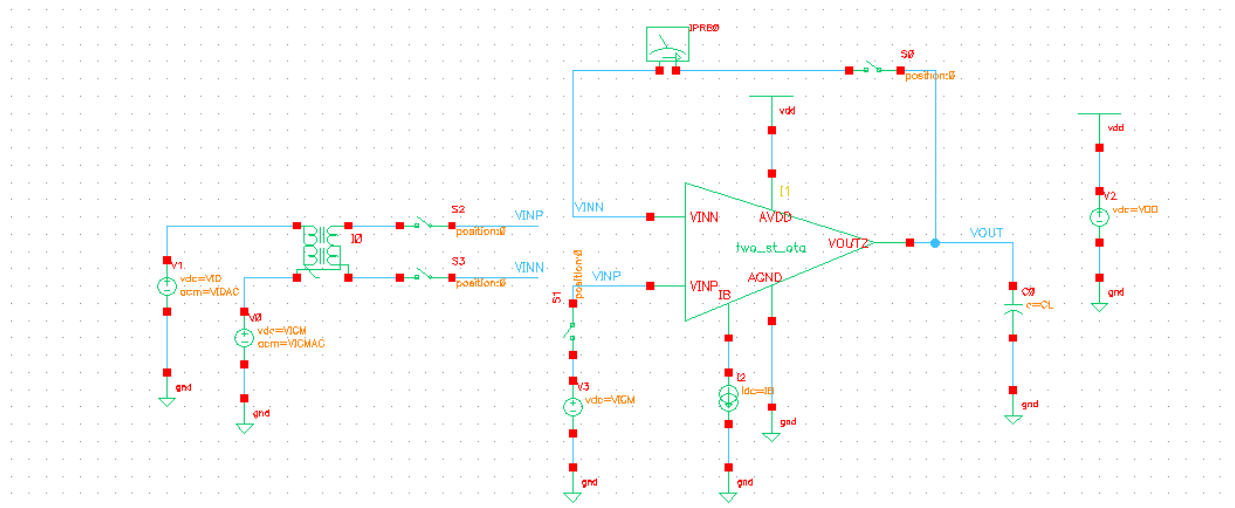


Fig.26: DC CL AC OL Testbench

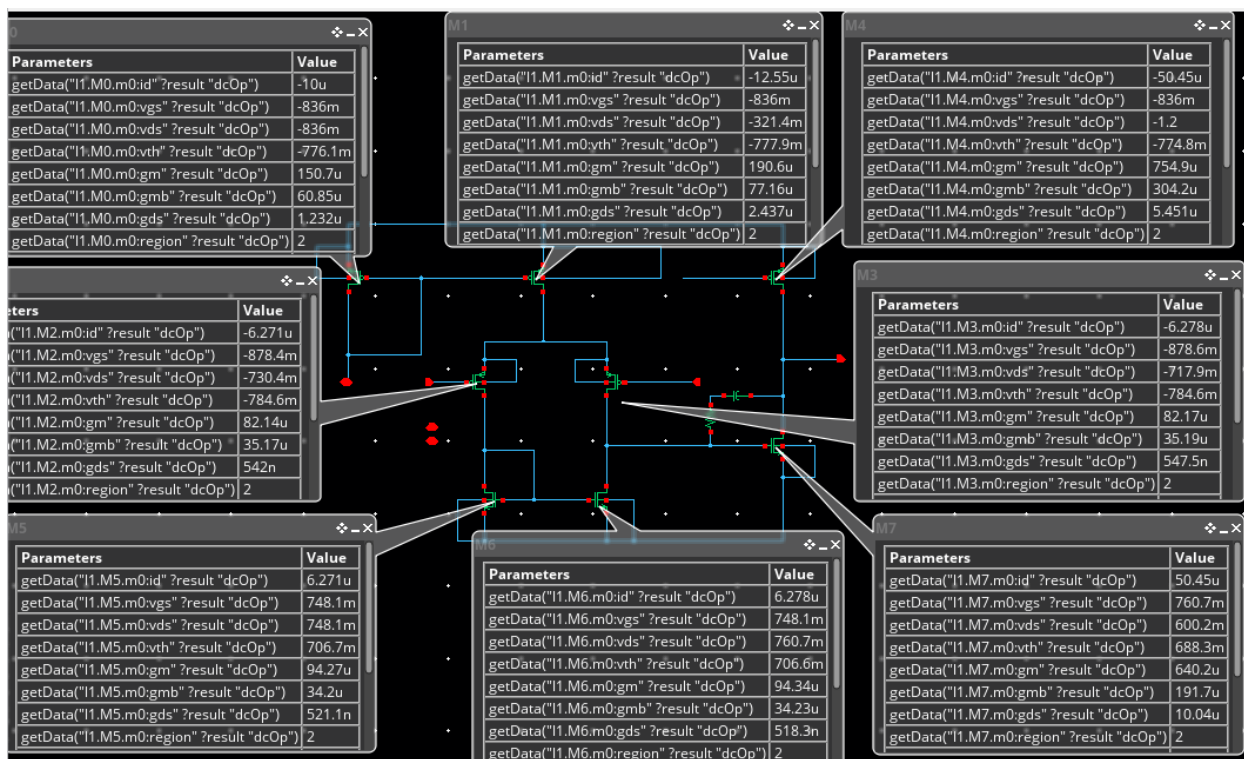


Fig.27: CL DC OL AC OTA OP

- All the devices are in saturation
- The change in g_m and I_D is (0.03%) is due to the amplifier's very high open-loop gain which minimizes the error across the inputs.

Diff. Small Signal Ccs

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade) $V_{in,diff,AC} = 1V, V_{in,CM} = 0.6V$

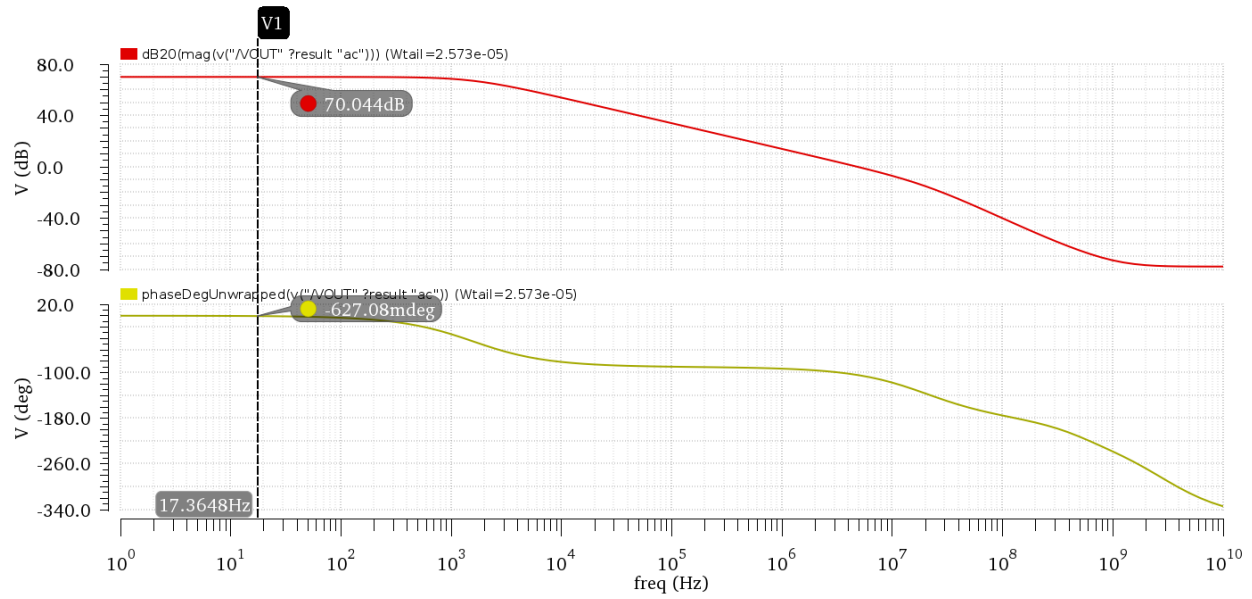


Fig.28: CL DC OL AC OTA Bode Plot of diff. gain mag. and phase

- Hand Analysis

$$A_{vd1} = g_{m2,3} R_{out1} = g_{m2,3}(r_{o3}/r_{o6})$$

$$A_{vd2} = g_{m7} R_{out2} = g_{m7}(r_{o4}/r_{o7})$$

$$A_{vd} = A_{vd1} \times A_{vd2}$$

$$\omega_{p1} \approx \frac{1}{R_{out1}(G_{m2}R_{out2})C_c}, \quad BW = \frac{\omega_{p1}}{2\pi}$$

$$GBW = \frac{g_{m2,3}}{C_c}$$

$$UGF = \omega_u \approx GBW = A_{vd} \times BW$$

Parameter	Analytical	Simulation
Diff. gain A_{vd}	3.185k	3.175k
Diff. gain A_{vd} dB	70.06 dB	70.04 dB
BW	1.64 kHz	1.583 kHz
GBW	5.23 MHz	5.044 MHz
UGF	5.23 MHz	5.032 MHz

Part 5: DC Closed-Loop AC Open-Loop OTA Simulation

Ao	3.179k	> 2k		pass
Ao dB	70.04	> 66		pass
BW	1.583k			
GBW	5.044M	> 5M		pass
fu	5.032M	> 5M		pass
CMRR				
CMRR dB	80.41	> 74		pass

Fig.29: Diff. Simulation Results

CM Small Signal Ccs

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade) $V_{inCMAC} = 1V$, $V_{inCM} = 0.6V$

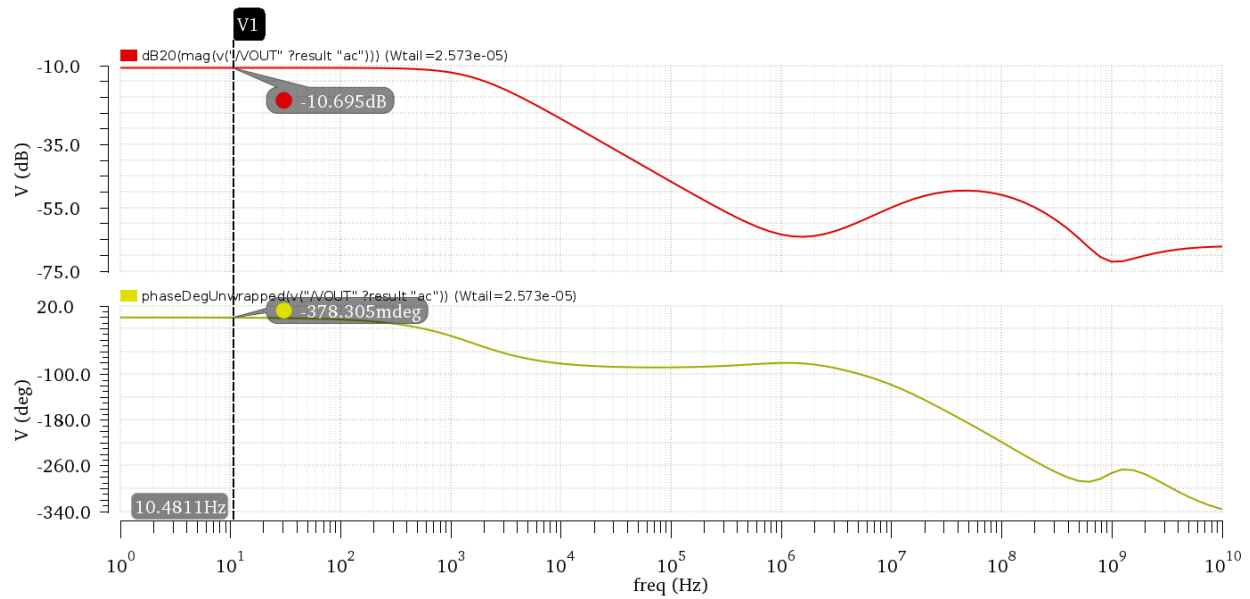


Fig.30: CL DC OL AC OTA Bode Plot of CM gain mag. and phase

- Hand Analysis

$$A_{vCM_1} = \frac{1}{2g_{m5,6}R_{ss}} = \frac{g_{ds1}}{2g_{m5,6}}$$

$$A_{vCM} = A_{vCM_1} \times A_{vd_2}$$

Parameter	Analytical	Simulation
CM gain A_{vCM}	533.8m	291.9m
CM gain A_{vCM} dB	-5.46 dB	-10.69 dB
CMRR	75.53 dB	80.74 dB

Part 5: DC Closed-Loop AC Open-Loop OTA Simulation

Ao	291.9m	< 400m		pass
Ao dB	-10.69	< -7.97		pass
CMRR				
CMRR dB	80.74	> 74		pass

Fig.31: CM Simulation Results

CMRR

$$CMRR = \frac{A_{vd}}{A_{vCM}}$$

$$CMRR_{dB} = A_{vd_{dB}} - A_{vCM_{dB}}$$

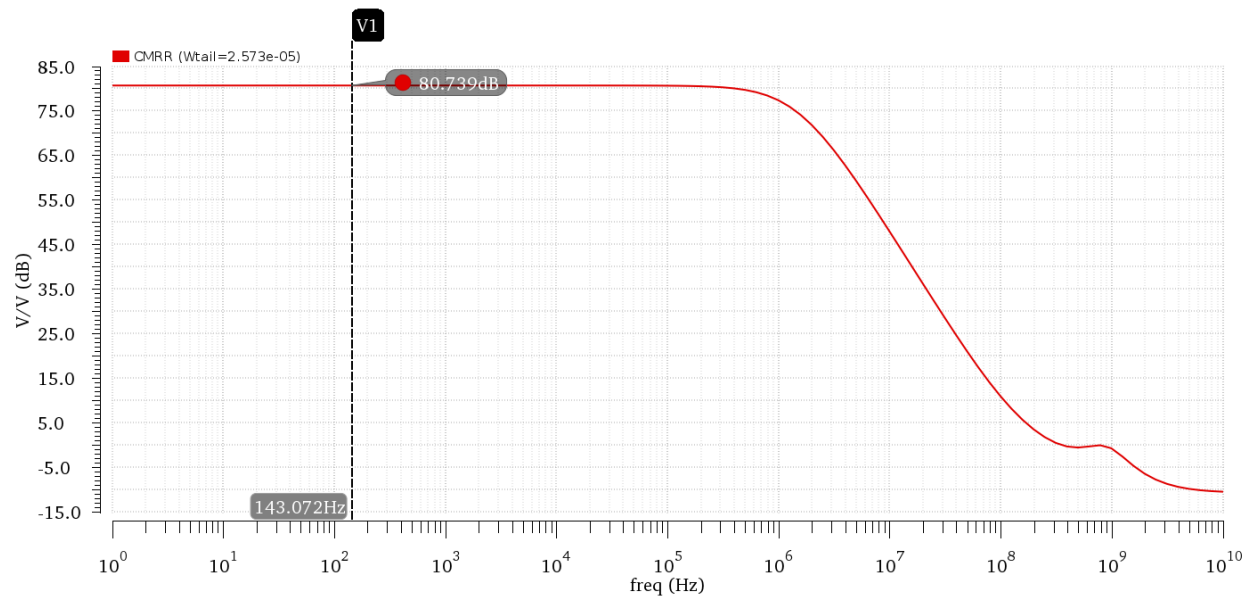


Fig.32: Bode Plot of CMRR

CM Large Signal Ccs (GBW vs V_{inCM})

Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade). $V_{in_{diff.AC}} = 1$ and $V_{in_{CM.AC}} = 0$

Use Parametric Sweep $V_{in_{CM}} = 0: 25m: V_{DD}$

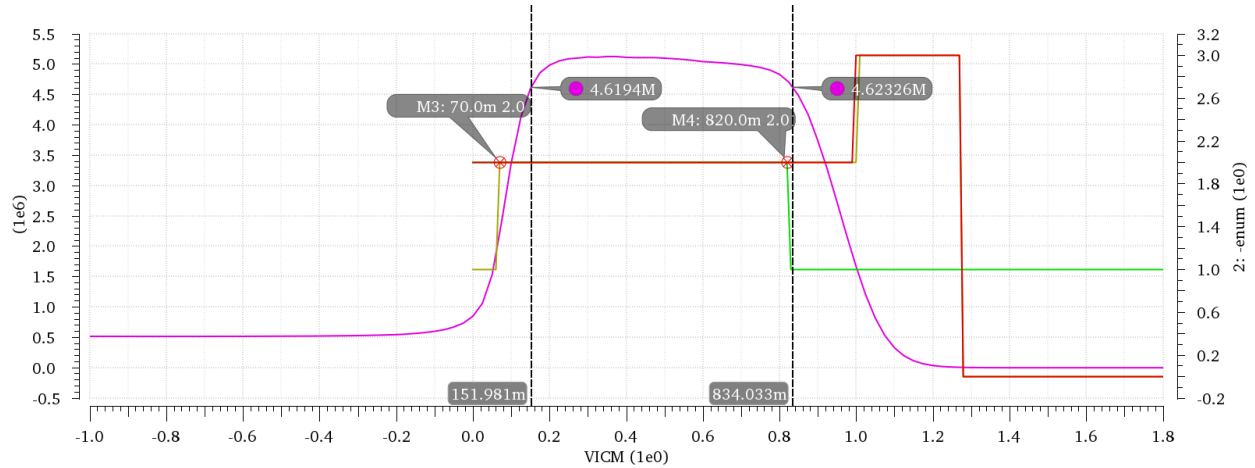


Fig.33: GBW vs V_{inCM}

$$GBW_{max} = 5.11 \text{ MHz}$$

$$90\% \text{ of } GBW = 4.61 \text{ MHz}$$

Parameter	region	OP	GBW
$CMIR_{Low}$	70mV	-36.4 mV	151.98 mV
$CMIR_{High}$	834 mV	863.5 mV	834 mV

We Noticed that the CMIR range extends to ground, due to body effect. In CMOS fabrication process PMOS is placed in n-well, so we can connect the source with body without any problem, source voltage drops within some range this affects the V_{SB} and make it decrease then the $|V_{THP}|$ magnitude increase so it pushes the minimum required gate voltage boundary lower and lower, effectively allowing CMIR to stretch all the way down to ground.

Design Specs

<i>Parameter</i>	<i>Required</i>	<i>Achieved</i>
<i>Technology</i>		0.18 μ
<i>Supply Voltage V_{DD}</i>	1.8V	1.8V
<i>Static Gain Error ϵ_r</i>	$\leq 0.05\%$	0.03%
<i>CMRR @ DC</i>	$\geq 74 \text{ dB}$	75.74 dB
<i>Phase Margin (avoid pole-zero doublets)</i>	$\geq 70^\circ$	73.84°
<i>OTA Current Consumption</i>	$\leq 60 \mu\text{A}$	59.94 μA
<i>CMIR – high</i>	$\geq 0.8V$	827.72 mV
<i>CMIR – low</i>	≤ 0.2	18.36 mV
<i>Output Swing</i>	0.2 – 1.6V	120.8mV – 1.691V
<i>Load C_L</i>	5pF	5pF
<i>Buffer Closed Loop rise time t_{rise}</i>	$\leq 70\text{ns}$	39.8 ns
<i>Slew Rate SR</i>	5 V/ μs	5.124 V/ μs

Table 3: Design Specs.